Vietnam International Mathematics Competition 2025

Danang City, 14th to 19th August 2025

Elementary Mathematics International Contest

Team Contest

Time limit: 70 minutes

Information:

- You are allowed 70 minutes for this paper, consisting of 10 questions printed on separate sheets. For questions 1, 3, 5, 7 and 9, only numerical answers are required. For questions 2, 4, 6, 8 and 10, full solutions are required.
- Each question is worth 40 points. For odd-numbered questions, no partial credits are given. There are no penalties for incorrect answers, but you must not give more than the number of answers being asked for. For questions asking for several answers, full credit will only be given if all correct answers are found. For even-numbered questions, partial credits may be awarded.
- Diagrams shown may not be drawn to scale.

Instructions:

- Write down your team's name in the space provided on every question sheet.
- Enter your answers at the bottom right corners of the corresponding question sheets.
- During the first 10 minutes, the four team members examine the first 8 questions together and discuss them. After 10 minutes they distribute the questions among themselves, with every team member allotted at least 1 question.
- During the next 35 minutes, the four team members write down the solutions of their allotted problems on the respective question sheets, with no further communication or discussion.
- During the last 25 minutes, the four team members work together to write down the solutions of the last 2 questions on the respective question sheets.
- It is forbidden to use instruments such as protractors, calculators and electronic devices, smart watches.
- At the end of the contest, you must hand in the envelope containing all question sheets and all scrap paper.

English Version

Team: _____

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Team Contest

16th August, 2025, Danang City, Vietnam

Team : _____ Solver : _____ ID : _____

1. A three-digit number $\overline{abc}$ is said to be *remarkable* if $\overline{abc}$ is less than 500 and the last three digits of $\overline{abc}^3$ are $\overline{abc}$. What is the sum of the largest and smallest possible *remarkable* numbers?

Answer: _____

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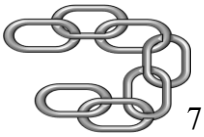
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2. Below, the left diagram shows a chain that has two connected rings and the right diagram shows a chain that has seven connected rings.



2 connected rings



7 connected rings

Mr. Metal owns 10 chains. Two of them are identical, each consisting of a single ring. The remaining eight chains have 2, 3, 5, 8, 13, 21, 34 and 55 connected rings, respectively, as shown below.



single ring



8 connected rings



13 connected rings



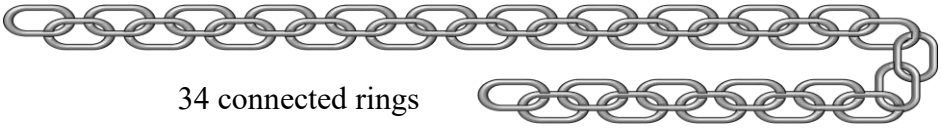
single ring



21 connected rings



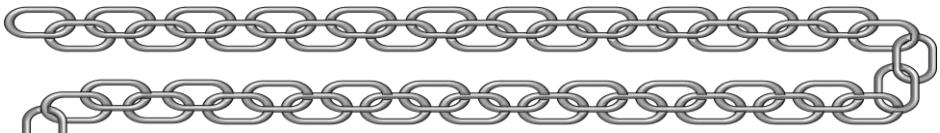
2 connected rings



34 connected rings

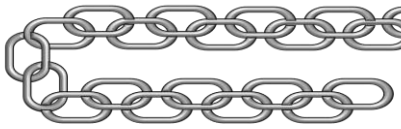


3 connected rings





5 connected rings



55 connected rings

Mr. Metal wants to make two or more chains, all of the same length, with no loops or any opened ring in it. All rings must be used. What is the shortest amount of time, in minutes, to make them if it takes exactly 1 minute to cut a ring (which disconnects it from other rings) and exactly 3 minutes to connect rings by welding one open ring (after connecting it to zero or more other rings)?

Example of Cutting Process A:



Takes exactly 1 minute

Example of Connecting Process A:



Takes exactly 3 minutes

Example of Cutting Process B:



Takes exactly 1 minute

Example of Connecting Process B:



Takes exactly 3 minutes

(Please write down your answer on the back)



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3. Let n be a positive integer that is a multiple of 18 and has exactly 15 positive divisors. What is the sum of all possible values of n?

Answer: _____

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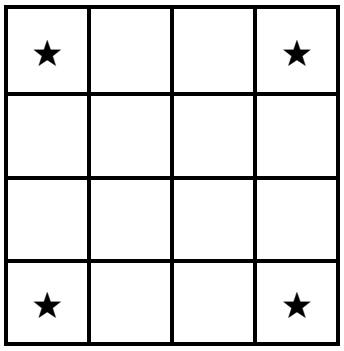
Elementary Mathematics International Contest

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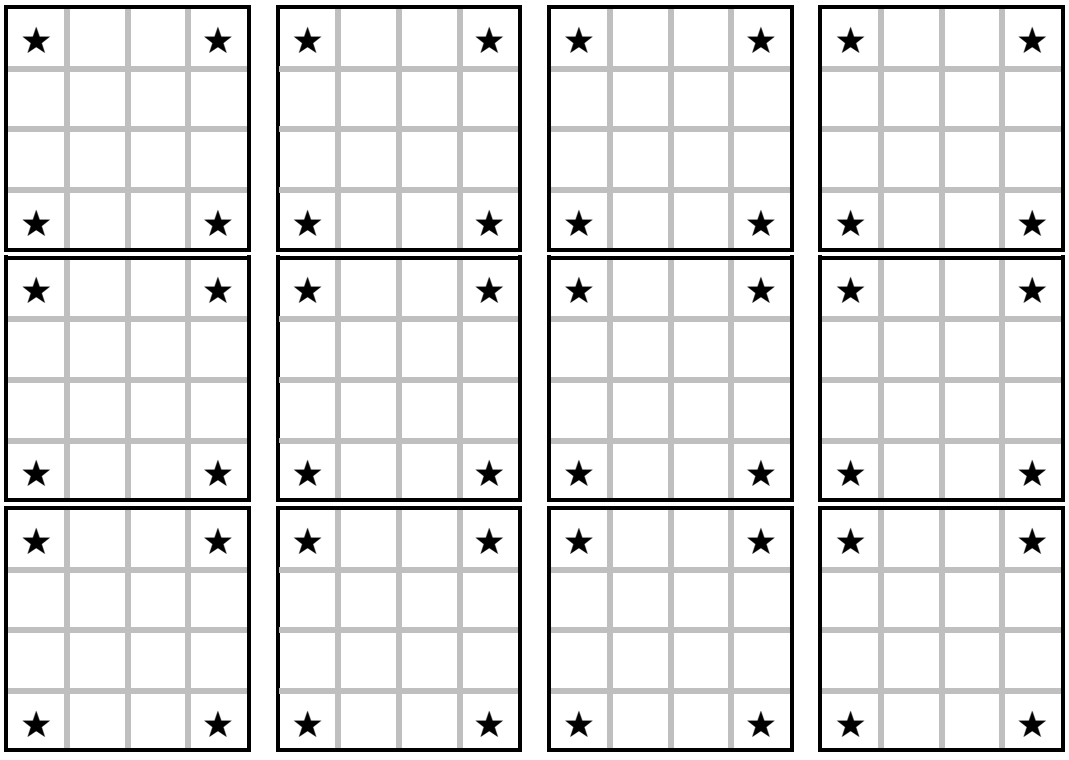
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4. The diagram below shows a 4×4 board made of 16 unit squares, which contains one star in each corner cell. The board is to be partitioned - by cutting along the grid lines - into non-overlapping regions, where each region is connected by sides. A partition is said to be valid if the regions all have the same area and contain exactly one star each. Draw all the valid partitions of the board. (Note: Two partitions are considered the same if they can be obtained from each other by a finite number of reflections and/or rotations.)



Answer:



(Note: You may not need to fill out all the provided answer diagrams.)

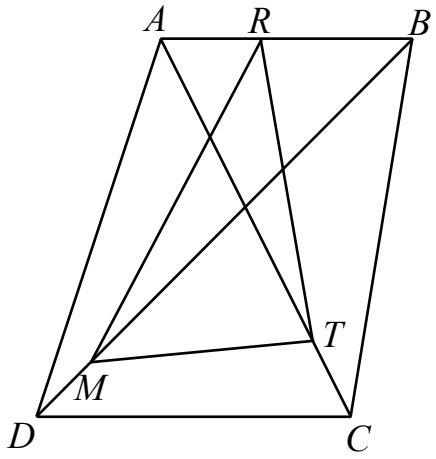
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5. In the diagram below, $ABCD$ is a quadrilateral of area 675 cm^2, where AB and CD are parallel and $AB = 20$ cm, $CD = 25$ cm. Points M, R and T are on segments BD, AB and AC, respectively, such that $\frac{MD}{MB} = \frac{1}{6}$ and $\frac{TC}{TA} = \frac{1}{4}$. If the area of the triangle MRT is 216 cm^2, what is the length, in cm, of AR?



Answer: _____ cm

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6. Three boxes, V, N and M, initially contain 20, 25 and 60 identical marbles, respectively. Two players take turns playing a game. On each turn, a player chooses one box and removes a positive number of marbles from it. However, the following conditions must always remain satisfied after each move:

- The number of marbles in box V must never exceed that in box N;
- The number of marbles in box N must never exceed that in box M.

The player who takes the last marble wins. One of the players has a winning strategy. Formulate a winning strategy for this player and prove that is works.

Answer: _____



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7. How many positive multiples of 2025×10101 have exactly 2025 positive divisors?

Answer: _____

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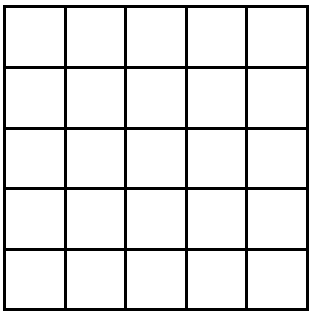
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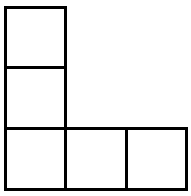
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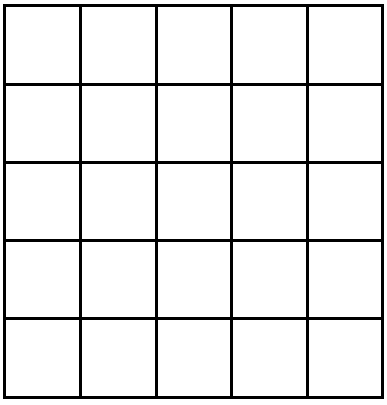
8. In a 5×5 board, each cell is to be painted with one of many colours.



What is the minimum number of distinct colours needed so that no matter how an L-shaped tile (shown below) is placed on the 5×5 board along its grid lines, it will always cover 5 cells of distinct colours? (Note: You must provide a colouring that satisfies your answer, and prove that using fewer colours is impossible.)



Answer: _____ colours,



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9. In a lottery game, a player chooses **6 distinct positive integers** from **1 to 42**.

A sample of the lottery ticket is shown below:

○ 1	○ 2	○ 3	○ 4	○ 5	○ 6	○ 7
○ 8	○ 9	○ 10	○ 11	○ 12	○ 13	○ 14
○ 15	○ 16	○ 17	○ 18	○ 19	○ 20	○ 21
○ 22	○ 23	○ 24	○ 25	○ 26	○ 27	○ 28
○ 29	○ 30	○ 31	○ 32	○ 33	○ 34	○ 35
○ 36	○ 37	○ 38	○ 39	○ 40	○ 41	○ 42

Now, Alex considers a ticket *perfect* if all the digits from 1 to 9 are used exactly once in the selected numbers, and the digit 0 is not used at all. For example, he can select 7, 8, 9, 14, 25 and 36, as shown in the diagram below:

○ 1	○ 2	○ 3	○ 4	○ 5	○ 6	● 7
● 8	● 9	○ 10	○ 11	○ 12	○ 13	● 14
○ 15	○ 16	○ 17	○ 18	○ 19	○ 20	○ 21
○ 22	○ 23	○ 24	● 25	○ 26	○ 27	○ 28
○ 29	○ 30	○ 31	○ 32	○ 33	○ 34	○ 35
● 36	○ 37	○ 38	○ 39	○ 40	○ 41	○ 42

In how many ways can he fill out a *perfect* ticket?

Answer: _____ ways

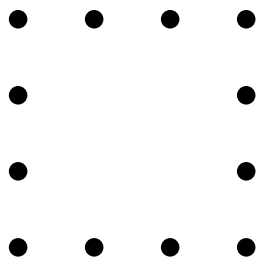
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10. The diagram below shows an evenly spaced 4×4 grid with the four central points removed. What is the total number of distinct circles that pass through at least 3 of the grid points shown?



Answer: _____ circles